Adithya Vadivel

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Education

University of Maryland Sept 2025 – May 2028

Bachelor of Science, Computer Science and Data Science

- **GPA:** 3.7/4.0
- **Organizations:** College Park Scholars (Science, Technology, and Society)
- **Achievements:** Academic Honors, Dean's List
- **Coursework:** Object Oriented Programming, Data Structures and Algorithms, Discrete Structures, Databases

Technical Stack

- **Programming Languages:** Java Kotlin, Python, MATLAB, C/C++, HTML, CSS, JavaScript, SQL
- **Frameworks:** React.js, Vite, Firebase, Git
- **Certifications:** AWS Certified Cloud Practitioner, PCEP – Certified Entry-Level Python Programmer

Experience

Northeastern University – Center for Drug Discovery Oct 2024 – Dec 2024

Research Intern (10 hrs/week) Boston, MA

- Conducted literature reviews on positive and negative effects of cannabinoids on public health, collaborated with mentors and peers to refine research scope, and escalated findings and questions to senior staff.
- Utilized ChemDraw, SciFinder, PubMed, EndNote, and RePORTER to collect, organize, and analyze research data, improving efficiency in data management and retrieval.
- Leveraged SQL-based data organization and querying to manage and retrieve structured research information, improving efficiency in data access and analysis.
- Applied analytical problem-solving and debugging techniques when working with complex datasets and research tools.

Aegle Nutrition Jul 2024 – Oct 2024

Quality Control Intern (10 hrs/week) Carrollton, TX

- Acquired knowledge on code of federal regulations (21CFR 117) for Current Good Manufacturing Practices (cGMP) and real-time life cycle management of dietary supplements, and tracked process documentation.
- Worked with regulated data pipelines by tracking, validating, and managing cGMP (21 CFR 117) process documentation across workflows.

Projects

Personal Portfolio 2026

- Developed and deployed a responsive single-page web application using React + Vite, focusing on component-based architecture and clean UI design.
- Implemented cloud hosting and CDN delivery using AWS S3 and CloudFront, with custom domain routing via Route 53 and HTTPS enabled through AWS Certificate Manager.
- Optimized load performance and reliability through static asset bundling, caching, and secure deployment practices.

Media Rental Manager 2025

CMSC131 (Object Oriented Programming)

- Designed and implemented a Java-based media rental system, applying object-oriented principles such as encapsulation, inheritance, and polymorphism to ensure modular and scalable design.
- Integrated RESTful design principles and optimized performance via caching and asset bundling
- Emphasized modular design, separation of concerns, and clean APIs to improve maintainability and scalability.
- Optimized data access patterns using efficient data structures, improving lookup and rental processing performance.

Passport Project 2025

CMSC131 (Object Oriented Programming)

- Developed a Java-based domain model using encapsulation and data validation, ensuring consistency of object state across operations.
- Wrote JUnit test cases to validate edge cases and core functionality, improving reliability and reducing logical errors.
- Applied systematic debugging techniques to identify and resolve logical inconsistencies in object behavior